

Patient-reported outcomes in a phase 2 study comparing atezolizumab alone or with bevacizumab vs sunitinib in previously untreated metastatic renal cell carcinoma

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Objective

To evaluate patient-reported outcome (PRO) data from the IMmotion150 study. The phase 2 IMmotion150 study showed improved progression-free survival with atezolizumab plus bevacizumab vs sunitinib in patients with programmed death-ligand 1 (PD-L1)+ tumours and suggested activity of atezolizumab monotherapy in previously untreated metastatic renal cell carcinoma (mRCC).

Patients and methods

Patients with previously untreated mRCC were randomised to atezolizumab 1200 mg intravenously (i.v.) every 3 weeks ($n = 103$), the atezolizumab regimen plus bevacizumab 15 mg/kg i.v. every 3 weeks ($n = 101$), or sunitinib 50 mg orally daily (4 weeks on, 2 weeks off; $n = 101$). The MD Anderson Symptom Inventory (MDASI) and Brief Fatigue Inventory (BFI) were administered on days 1 and 22 of each 6-week cycle. Time to deterioration (TTD), change from baseline in MDASI core and RCC symptom severity, interference with daily life, and BFI fatigue severity and interference scores were reported for all comers. The TTD was the first ≥ 2 -point score increase over baseline. Absolute effect size ≥ 0.2 suggested a clinically important difference with checkpoint inhibitor therapy vs sunitinib.

Results

Completion rates were $>90\%$ at baseline and $\geq 80\%$ at most visits. Delayed TTD in core and RCC symptoms, symptom interference, fatigue, and fatigue-related interference was observed with atezolizumab (both alone and in combination) vs sunitinib. Improved TTD (hazard ratio [HR], 95% confidence interval [CI]) was more pronounced with atezolizumab monotherapy: core symptoms, 0.39 (0.22–0.71); RCC symptoms, 0.22 (0.12–0.41); and symptom interference, 0.36 (0.22–0.58). Change from baseline by visit, evaluated by the MDASI, also showed a trend favouring atezolizumab monotherapy vs sunitinib. Small sample sizes may have limited the ability to draw definitive conclusions.

Conclusion

PROs suggested that atezolizumab alone or with bevacizumab maintained daily function compared with sunitinib. Notably, symptoms were least severe with atezolizumab alone vs sunitinib (IMmotion150; ClinicalTrials.gov Identifier: NCT01984242).

Keywords

immunotherapy, patient-reported outcomes, quality of life, atezolizumab, bevacizumab, IMmotion150

Introduction

Vascular endothelial growth factor (VEGF) inhibition with VEGF monoclonal antibodies, such as bevacizumab, or receptor tyrosine kinase inhibitors (VEGFR-TKIs), such as sunitinib or pazopanib, has been a first-line standard for over a decade in metastatic renal cell carcinoma (mRCC) due to the angiogenic nature of the tumour [1]. However, this paradigm has changed on the basis of multiple studies assessing innovative combinations of either dual checkpoint inhibitor (CPI) therapy or VEGF-directed agents with CPI. The programmed cell death 1 protein (PD-1) inhibitor nivolumab, when combined with the cytotoxic T-lymphocyte-associated protein 4 inhibitor ipilimumab, has demonstrated improved survival and response rate relative to sunitinib in patients with mRCC and intermediate- or poor-risk prognostic features [2]. The VEGFR-TKI axitinib has been combined with pembrolizumab (a PD-1 inhibitor) and avelumab (a programmed death-ligand 1 [PD-L1] inhibitor), with improved clinical outcomes over sunitinib with both combinations [3,4]. The VEGF monoclonal antibody bevacizumab combined with the PD-L1 inhibitor atezolizumab has similarly shown clinical benefit and shown prolonged progression-free survival (PFS) relative to sunitinib in a phase 3 study [5].

The efficacy of CPI combinations in mRCC gives rise to the question of whether CPI monotherapy could also represent an up-front approach. To this end, the non-randomised, prospective KEYNOTE-427 study (ClinicalTrials.gov Identifier: NCT02853344) examined 110 patients with metastatic clear cell RCC receiving first-line pembrolizumab [6]. A response rate of 36.4% (95% CI 27.4–46.1%) was seen in 110 evaluable patients, and the median (95% CI) PFS was 7.1 (5.6–11.0) months. We further evaluated CPI monotherapy in the IMmotion150 study (ClinicalTrials.gov Identifier: NCT01984242), a randomised phase 2 study, to assess the safety and efficacy of atezolizumab monotherapy or atezolizumab plus bevacizumab vs sunitinib in 305 patients with treatment-naïve mRCC [5]. Objective response rates were 25%, 32%, and 29% with atezolizumab monotherapy, atezolizumab plus bevacizumab, and sunitinib, respectively, in the intention-to-treat (ITT) population, and 28%, 46%, and 27%, respectively, in the PD-L1+ population [7].

The results of the IMmotion150 study have been further examined in the IMmotion151 study (ClinicalTrials.gov Identifier: NCT02420821). The IMmotion151 study is a randomised, open-label, phase 3 study comparing atezolizumab plus bevacizumab vs sunitinib in previously untreated mRCC. The median PFS was 11.2 months with atezolizumab plus bevacizumab vs 7.7 months with sunitinib in the PD-L1+ patient population [hazard ratio (HR) 0.74, 95% CI 0.57–0.96; $P = 0.022$] [5]. Patient-reported outcome (PRO) data from the IMmotion151 study favoured

atezolizumab plus bevacizumab vs sunitinib. The median time to clinically relevant deterioration of daily functioning due to treatment- and disease-related symptoms was notably longer in patients receiving atezolizumab plus bevacizumab (11.3 months) than in patients receiving sunitinib (4.3 months) in the ITT population (stratified HR 0.56, 95% CI 0.46–0.68).

IMmotion150 presents a rare opportunity to juxtapose CPI monotherapy simultaneously against VEGFR-TKI and combination strategies. Because patients with cancer are burdened by both disease- and treatment-related symptoms, the assessment and lessening of these symptoms is important in therapeutic decision making. In the present study, we compare PRO data for patients randomly assigned to one of these three distinct treatment approaches.

Patients and methods

Study design and eligibility

IMmotion150 is a multicentre, prospective, randomised, open-label, phase 2 study conducted at 96 institutions. This hypothesis-generating study was designed to evaluate the safety and preliminary activity of atezolizumab plus bevacizumab vs sunitinib and atezolizumab monotherapy vs sunitinib. The study was designed to enrol ~100 patients per arm regardless of PD-L1 status to enable adequate subset analysis. Patients eligible for the study had mRCC with clear cell and/or sarcomatoid components. No prior systemic therapy was allowed. Patients with active CNS disease were excluded, and patients had to have adequate haematological and laboratory parameters [7].

Patients completed PRO assessments before initiating treatment (i.e., at baseline), on days 1 and 22 of each 6-week treatment cycle, and at the end of treatment. Given the two different dosing schedules, patients in the sunitinib arm were off treatment for 2 weeks at day 1 assessment visits and were receiving their 28-day treatment at day 22 assessment visits.

Treatment regimen

Patients were randomised to atezolizumab 1200 mg intravenously (i.v.) every 3 weeks alone or with bevacizumab 15 mg/kg i.v. every 3 weeks, or sunitinib 50 mg oral daily (4 weeks on, 2 weeks off). Stratification factors included Memorial Sloan Kettering Cancer Center (MSKCC) risk category (low, intermediate, or high), prior nephrectomy (yes or no), and PD-L1 status ($\geq 5\%$ or $< 5\%$). Patients were treated until disease progression or unacceptable toxicity. Notably, patients who derived clinical benefit despite disease progression were allowed to continue study treatment with atezolizumab. The study allowed for crossover from sunitinib and atezolizumab monotherapy to combination therapy. However,

crossover following atezolizumab therapy was not allowed in the European centres due to local regulatory guidance.

PRO instruments

MD Anderson Symptom Inventory (MDASI)

The MDASI has been validated among patients diagnosed with various cancer types and undergoing different forms of treatment [8]. It comprises 13 core items that assess severity of disease and treatment-related symptoms [6–8]. An additional three items were used to assess RCC-specific symptom severity (mouth/throat sores, rash/skin change, and headache). Patients were asked to rate each symptom at its worst in the past 24 h on a scale from zero to 10, with 10 representing the worst imaginable severity. A further six items assessed symptom interference with daily life on the same 11-point scale, with 10 representing complete interference with daily life. The six dimensions of daily life include: general activity, walking, work, mood, relations with other people, and enjoyment of life.

Brief Fatigue Inventory (BFI)

The BFI is a nine-item instrument designed to assess patient-reported cancer-related fatigue [9]. BFI has been found to be unidimensional and to be a valid and reliable measure [9]. A single item, instead of three items, was used to assess the severity of fatigue at its worst in the past 24 h on a scale from zero to 10, with 10 indicating the worst severity. Further questions assessed fatigue-related interference across the six dimensions of daily life noted in the previous section.

Statistical analysis

The clinical data cutoff date was 19 April 2017. PROs were pre-specified to evaluate disease- and treatment-related symptom severity (including fatigue) and their interference with patients' daily functioning. Analyses included PRO data collected up to crossover for patients randomised to the atezolizumab monotherapy or sunitinib arms, and up to the end of treatment for patients randomised to the atezolizumab plus bevacizumab arm or patients randomised to the atezolizumab monotherapy arm in Europe. PROs were analysed without type I error control. Completion rates were calculated as the number of patients who completed assessments divided by the number of patients expected to complete assessments at each visit. Per developers' user manuals, for PRO scales with >50% of the constituent items completed, a prorated score was computed. For scales with ≤50% of the items completed, the scale was considered missing.

Longitudinal analysis of change in each PRO score from baseline included scheduled on-study visits among all

randomised patients with a non-missing baseline score and at least one post-baseline score. The primary longitudinal analysis was based on repeated-measures models including: baseline PRO score, stratification factors, treatment, visit, and treatment-by-visit interaction. Supportive analysis based on linear mixed-effects models included: baseline PRO score, stratification factors, treatment, and time (continuous) as fixed effects as well as random effects of intercept and slope (for time). Least-squares mean (LSM) change from baseline within each treatment arm and the difference in the LSM change between treatment arms were estimated at each visit (repeated-measures models) or over the first-line treatment period (linear models). Repeated-measures model estimates are reported up to the last consecutive visit when ≥20% of patients were still on study. Effect size (ES) supported interpretation of differences between treatment arms, where absolute ES ≥0.20 likely represented a clinically important difference [10]. The ES was calculated as the difference in score change divided by the pooled standard deviation at each visit. Negative ES values favoured atezolizumab alone or atezolizumab plus bevacizumab vs sunitinib, whereas positive ES values favoured sunitinib.

Time-to-event analyses were evaluated as the time to clinically meaningful deterioration, defined as the time from randomisation date to a patient's first ≥2-point score increase above baseline. The HRs and 95% CIs by treatment arm were estimated using a stratified Cox regression model. Kaplan–Meier methodology was used to estimate the probability of deterioration. Analyses were performed on all randomised patients. Patients without a deterioration event were censored at the last non-missing PRO assessment date, whereas patients with no baseline or post-baseline PRO assessment were censored at randomisation.

Results

Patient characteristics

The IMmotion150 study enrolled 305 patients between 8 January 2014 and 16 March 2015. Of those enrolled, 101 patients were randomised to the atezolizumab plus bevacizumab arm, 103 to the atezolizumab monotherapy arm, and 101 to the sunitinib arm. As previously reported, baseline patient and tumour characteristics were generally well balanced across treatment arms (Table S1) [7].

PRO scores

Completion rates for MDASI and BFI were >90% at baseline and ≥80% at most visits across arms. Baseline PRO scores indicating mild symptoms and interference were similar across arms (Table S2). There was a markedly improved time to deterioration in MDASI core symptom severity, RCC symptom severity, and symptom interference with

atezolizumab monotherapy vs sunitinib: core symptoms, HR (95% CI), 0.39 (0.22–0.71); RCC symptoms, 0.22 (0.12–0.41); and symptom interference, 0.36 (0.22–0.58; Fig. 1A–C). There were similar trends with the combination of atezolizumab plus bevacizumab vs sunitinib, although the differences were less pronounced: core symptoms, HR (95% CI), 0.74 (0.45–1.20); RCC symptoms, 0.60 (0.38–0.94); and symptom interference, 0.70 (0.47–1.04). Atezolizumab monotherapy also compared favourably to sunitinib in deterioration-free rate using BFI fatigue severity and fatigue-related interference scales; the results also showed a trend in favour of atezolizumab plus bevacizumab vs sunitinib (Fig. 1D,E).

Differences in core symptom severity, RCC symptom severity, and symptom interference were generally clinically relevant, favouring atezolizumab monotherapy vs sunitinib (Fig. 2A–C). Linear mixed-effects model estimates were generally consistent with repeated-measures model estimates, suggesting that patients also reported a lesser degree of symptom severity and symptom interference with atezolizumab monotherapy vs sunitinib over the first-line treatment period. Differences in LSM change (95% CI) and corresponding ES for atezolizumab monotherapy vs sunitinib were as follows: core symptoms, -0.47 (-0.76 to -0.17) and -0.33 ; RCC symptoms, -0.64 (-0.94 to -0.33) and -0.70 ; and symptom interference, -0.80 (-1.24 to -0.36) and -0.36 . For atezolizumab plus bevacizumab vs sunitinib, changes in core symptom severity, RCC symptom severity, and symptom interference were similar. Differences in LSM change (95% CI) during first-line treatment and corresponding ES for atezolizumab plus bevacizumab vs sunitinib were as follows: core symptoms, -0.07 (-0.36 to 0.22) and -0.05 ; RCC symptoms, -0.15 (-0.45 to 0.16) and -0.20 ; and symptom interference, -0.28 (-0.71 to 0.15) and -0.13 .

All 16 symptoms assessed for severity were milder with atezolizumab vs sunitinib during first-line treatment (Fig. 3). The five symptoms with the largest increase in severity from baseline (dry mouth, fatigue, rash, drowsiness, and lack of appetite) were more prominently changed in the sunitinib arm relative to atezolizumab monotherapy or atezolizumab plus bevacizumab. The LSM change from baseline estimates by visit and over the first-line treatment period for BFI scales suggested nominally milder fatigue and less fatigue-related interference with atezolizumab monotherapy vs sunitinib, and similar changes in fatigue and fatigue-related interference with daily life with atezolizumab plus bevacizumab vs sunitinib.

Discussion

The present study results represent the first randomised PRO data directly comparing VEGFR-TKI therapy to CPI monotherapy in the frontline setting. Patients receiving atezolizumab monotherapy reported milder disease- and

treatment-related symptoms (including fatigue), delayed symptom deterioration, and less interference of symptoms with daily life. The collective results imply that characterising the subset of patients who benefit from atezolizumab monotherapy is of paramount importance, as this strategy could confer superior quality of life (QOL) relative to therapeutic alternatives. Of note, a less pronounced difference in QOL between atezolizumab plus bevacizumab and sunitinib was observed in the present study compared with the phase 3 IMmotion151 trial (previously reported) [11]. These differences could be due to a larger sample size in the phase 3 experience or to experiential gain with this novel doublet (or other anti-VEGF/CPI combinations) in the transition from the phase 2 to the phase 3 study.

With consistency across studies and disease states, monotherapy with PD-1/PD-L1-directed therapies has been associated with improved QOL relative to previous standards. Atezolizumab has been compared with docetaxel in patients with non-small cell lung cancer receiving second-line treatment [12]. Compared with docetaxel, atezolizumab delayed time to deterioration in physical function and led to a numerical improvement in health-related QOL (HRQOL). In mRCC, the only published randomised experience in which QOL was assessed with CPI monotherapy vs standard treatment is the CheckMate 025 trial (ClinicalTrials.gov Identifier: NCT01668784) [13]. In that trial, patients with prior VEGF-directed therapy were randomised to receive either nivolumab or the mechanistic target of rapamycin inhibitor everolimus. The HRQOL was assessed using the Functional Assessment of Cancer Therapy-Kidney Symptom Index-Disease Related Symptoms questionnaire. A clinically meaningful improvement in HRQOL was seen with nivolumab vs everolimus. Indeed, meta-analytic data pooling QOL data from randomised trials evaluating PD-1/PD-L1-directed therapies showed a global trend favouring these agents [14]. In mRCC, dual CPI therapy represents a current frontline standard based on the phase 3 CheckMate 214 trial (ClinicalTrials.gov Identifier: NCT02231749) [2]. That study demonstrated a significant improvement in survival in intermediate- and poor-risk patients with mRCC, and PRO data suggested improved HRQOL relative to sunitinib in this population [15]. With all of the caveats of cross-trial comparisons, a monotherapy approach may still have merit on the basis of multiple factors, including relatively similar efficacy and reduced clinical toxicity. A phase 3 study is planned comparing nivolumab plus ipilimumab to nivolumab plus placebo to address both the efficacy and toxicity impacts of a combination CPI vs a single agent.

CPI monotherapy is of particular interest in the adjuvant setting, where maintenance of QOL is essential. In melanoma, approvals exist for ipilimumab, nivolumab, and pembrolizumab in the adjuvant setting on the basis of separate, randomised, phase 3 trials [16–18]. Durvalumab is

Fig. 1 Time to deterioration based on (A) MDASI core symptom severity, (B) MDASI RCC symptom severity, (C) MDASI symptom interference, (D) BFI fatigue severity, and (E) BFI fatigue interference. Atezo, atezolizumab; bev, bevacizumab; BFI, Brief Fatigue Inventory; HR, hazard ratio; MDASI, MD Anderson Symptom Inventory; NE, not evaluable.

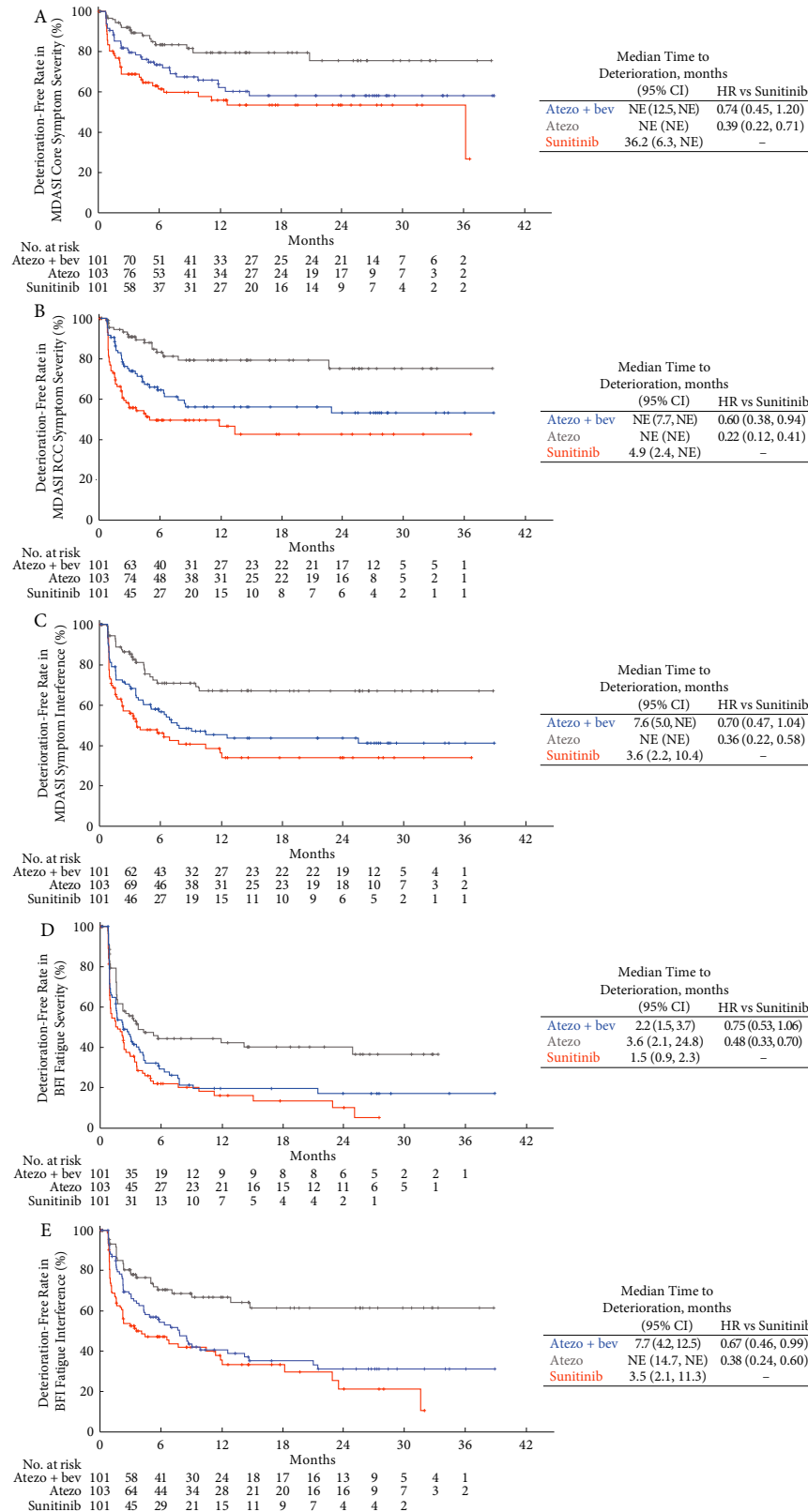


Fig. 2 Change from baseline by visit based on MDASI scales assessing (A) core symptom severity, (B) RCC symptom severity, and (C) symptom interference. Atezo, atezolizumab; bev, bevacizumab; ES, effect size; MDASI, MD Anderson Symptom Inventory.^a ES ≤ -0.20 favouring atezolizumab alone vs sunitinib. ^b ES ≤ -0.20 favouring atezolizumab plus bevacizumab vs sunitinib.

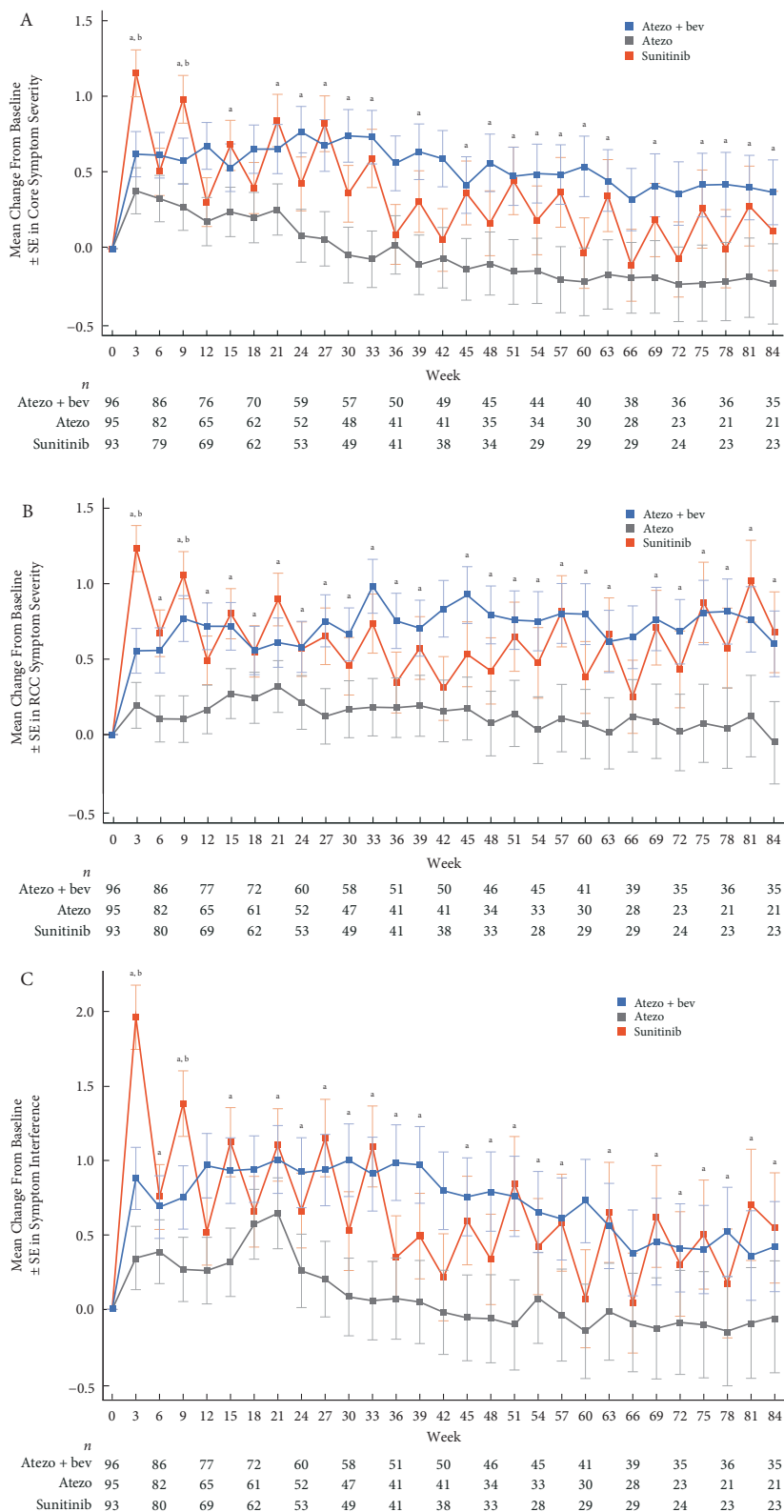
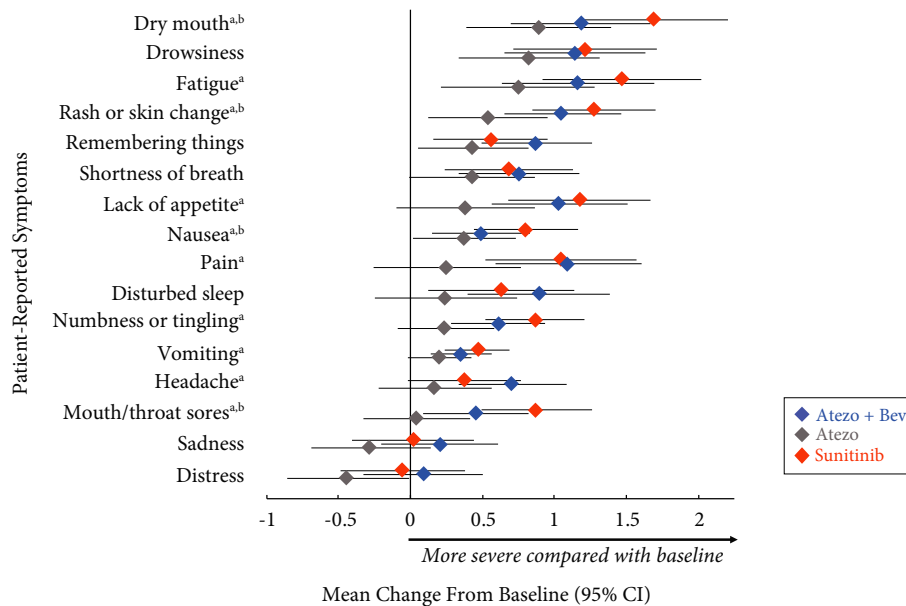


Fig. 3 Change in individual symptom severity during first-line treatment based on MDASI. Atezo, atezolizumab; bev, bevacizumab; ES, effect size; MDASI, MD Anderson Symptom Inventory. ^aES ≤ -0.20 favouring atezolizumab alone vs sunitinib. ^bES ≤ -0.20 favouring atezolizumab plus bevacizumab vs sunitinib.



approved for patients with Stage 3 non-small cell lung cancer, as consolidation treatment following combined chemoradiotherapy [19]. Following suit, several trials have been launched in patients with high-risk, localised RCC evaluating CPIs either before or after surgical resection. The IMmotion010 trial (ClinicalTrials.gov Identifier: NCT03024996) comparing atezolizumab to placebo in this setting has recently completed accrual, and a similar trial comparing pembrolizumab to placebo is currently underway [20,21]. An Eastern Cooperative Oncology Group (ECOG)-American College of Radiology Imaging Network (ACRIN)-led study, PROSPER (NCT03055013), is evaluating perioperative nivolumab therapy against observation [22]. Trials are evaluating dual CPIs in this setting as well [e.g., CheckMate 914 (NCT03138512), comparing nivolumab/ipilimumab to placebo). As all adjuvant studies use observation/placebo as the comparator arm, our present study may help to inform the treatment decision between sunitinib [currently approved as adjuvant therapy based on the phase 3 S-TRAC trial (ClinicalTrials.gov Identifier: NCT00375674) by the US Food and Drug Administration] and any emerging immunotherapy option in this setting [23]. Considerations around efficacy notwithstanding, our present data implied superior QOL with CPI monotherapy vs VEGFR-TKI therapy.

Although PRO questionnaire-completion rates were high throughout the study, results from this hypothesis-generating phase 2 study should be interpreted in the context of the

relatively small sample sizes. While PRO data did accompany the phase 3 IMmotion151 study, a much larger experience, our present study offers the unique and perhaps sole opportunity to juxtapose QOL with VEGFR-TKI therapy and CPI monotherapy [11].

During first-line treatment in IMmotion150, patients with mRCC treated with atezolizumab alone reported milder disease- and treatment-related symptoms (including fatigue), delayed symptom deterioration, and a lesser impact of symptoms on their daily life compared with those treated with sunitinib. Clinically relevant differences were most pronounced when comparing atezolizumab monotherapy with sunitinib. Although not statistically significant, a trend towards improved QOL was seen in patients treated with atezolizumab plus bevacizumab vs sunitinib. Patients treated with the combination reported similar symptom burden vs sunitinib. The QOL results were more compelling with atezolizumab monotherapy, suggesting that monotherapy should be explored in selected settings (e.g., biomarker-selected populations in a metastatic setting, or as adjuvant treatment where immunotherapy is under investigation and use of sunitinib is controversial due to its low risk-benefit ratio).

Acknowledgements

The authors thank Qian (Cindy) Zhu and Yong Wang for their contribution to statistical analysis. Patients treated at Memorial Sloan Kettering Cancer Center were supported in

part by Memorial Sloan Kettering Cancer Center Support Grant/Core Grant (P30 CA008748).

Conflict of interest

Sumanta K. Pal certifies that all conflicts of interest, including specific financial interests, relationships, and affiliations relevant to the subject matter or materials discussed in the manuscript (e.g., employment/affiliation, grants or funding, consultancies, honoraria, stock ownership or options, expert testimony, royalties, or patents filed, received, or pending), are the following: All authors report funding for editorial support from F. Hoffmann-La Roche, Ltd. Sumanta K. Pal has received consulting fees from Astellas, AVEO, Bristol-Myers Squibb (BMS), Eisai, Exelixis, Genentech, Ipsen, Novartis, F. Hoffmann-La Roche, Ltd, and Pfizer. David F. McDermott has received consulting fees from BMS, Pfizer, Merck, Novartis, Array BioPharma, Eli Lilly, EMD Serono, Jounce Therapeutics, Peloton, and Alkermes; and research grants from Prometheus Laboratories and BMS. Michael B. Atkins has received honoraria or fees for serving on advisory boards for BMS, Merck, Novartis, Arrowhead, Pfizer, Glactone, Werewolf, Fathom, Pneuma, and Leads; consulting fees from BMS, Merck, Novartis, Pfizer, Genentech/Roche, Exelixis, Eisai, AVEO, Array, AstraZeneca, Idera, Aduro, ImmunoCore, Boehringer, Ingelheim, Iovance, Newlink, Surface, Alexion, Acceleron, Cota, and Amgen; institutional support from BMS, Merck, Pfizer, and Genentech; and owns stock options in Werewolf and Pyxis Oncology. Bernard Escudier has received grants and honoraria from BMS and AVEO; grants from Novartis; and honoraria from Genentech/Roche, Pfizer, Oncorena, and Ipsen. Brian I. Rini has received grants and honoraria from Genentech/Roche and Pfizer; grants to his institution and honoraria for consulting roles from Merck, Peloton, AVEO, and BMA; grants to his institution from AstraZeneca; honoraria for consulting roles from Novartis, Synthorx, Compugen, Corvus, and Exelixis; and owns stock in PTC Therapeutics. Robert J. Motzer has received honoraria for consulting roles from BMS, Genentech/Roche, Pfizer, Novartis, Exelixis, Eisai, Incyte, Eli Lilly, and Merck, and institutional support from BMS, Genentech/Roche, Pfizer, Novartis, Exelixis, and Eisai. Lawrence Fong has received grants from Genentech/Roche, AbbVie, Bavarian Nordic, BMS, Dendreon, Janssen, and Merck. Richard W. Joseph has received honoraria for serving on advisory boards for BMS, Array, and Exelixis. Stephane Oudard has received honoraria for advisory roles from Novartis, MSD, BMS, Bayer, Pfizer, Ipsen, Genentech/Roche, Astellas, Sanofi, and Janssen. Alain Ravaud has received fees and non-financial support from Pfizer, AstraZeneca, BMS, Ipsen, Roche, and Novartis; and institutional support from Pfizer. Sergio Bracarda has received honoraria and travel support for advisory roles from Novartis, Astellas, Janssen, Pfizer, BMS, Roche, Bayer, and Ipsen; honoraria from Astellas

and Janssen and travel support from Exelixis and AstraZeneca. Cristina Suárez has received grants from Roche for conduct of the study; advisory board, speaking and travel fees from BMS and Pfizer; advisory board and speaking fees from Ipsen and Astellas; advisory board fees from Sanofi, Bayer, and Merck Sharp & Dohme; and travel expenses from Roche. Elaine T. Lam has received clinical trial funding from Genentech/Roche, Calithera, BMS, Peloton, Merck, and Pfizer. Toni K. Choueiri has received institutional support, grants, fees for consulting and advisory boards, and travel and accommodations from Pfizer, Exelixis AstraZeneca, Bayer, BMS, Cerulean, Eisai, Foundation Medicine, Genentech/Roche, GlaxoSmithKline, Merck, Novartis, Peloton, Prometheus Laboratories, Corvus, and Ipsen; institutional and research support from Tracoon; consulting and advisory board fees and travel and accommodations from Allegiant, UpToDate, National Comprehensive Cancer Network, Analysis Group, Michael J. Hennessy Associates, Inc, OncLive, Physicians' Education Resource, Lpath, *Kidney Cancer Journal*, Clinical Care Options, PlatformQ, Navinata Health, Harborside Press, American Society of Clinical Oncology, *The New England Journal of Medicine*, and *The Lancet Oncology*; research and institutional support from Calithera and Takeda; and holds patents related to two pending and issued patents on biomarkers predictive of immune checkpoint response. Beiyang Ding, Caroleen Quach, Kenji Hashimoto, and Elisabeth Piau-Louis are employees and hold stock options in Genentech/Roche. Christina Schiff was previously employed and previously owned stock options in Genentech/Roche. Thomas Powles has received research funding from AstraZeneca and Genentech/Roche; and honoraria and travel support from BMS, Ipsen, Exelixis, Roche, Merck, Pfizer, Novartis, AstraZeneca, Incyte, and Seattle Genetics.

Author contributions

All authors had full access to all the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. *Conception and design:* Sumanta K. Pal, David F. McDermott, Michael B. Atkins, Bernard Escudier, Brian I. Rini, Robert J. Motzer, Elisabeth Piau-Louis, Thomas Powles. *Acquisition of data:* Sumanta K. Pal, Michael B. Atkins, Brian I. Rini, Robert J. Motzer, Lawrence Fong, Richard W. Joseph, Stephane Oudard, Alain Ravaud, Sergio Bracarda, Cristina Suárez, Elaine T. Lam, Toni K. Choueiri, Kenji Hashimoto, Christina Schiff, Elisabeth Piau-Louis. *Analysis and interpretation of data:* Sumanta K. Pal, David F. McDermott, Michael B. Atkins, Bernard Escudier, Brian I. Rini, Robert J. Motzer, Richard W. Joseph, Stephane Oudard, Sergio Bracarda, Beiyang Ding, Caroleen Quach, Kenji Hashimoto, Christina Schiff, Elisabeth Piau-Louis, Thomas Powles. *Drafting of the manuscript:* Sumanta K. Pal, David F. McDermott, Bernard Escudier, Brian I. Rini, Robert

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Funding/support and role of the sponsor

F. Hoffmann-La Roche, Ltd sponsored the study and was involved in the design and conduct of the study; management, analysis, and interpretation of the data; and preparation, review, and approval of the manuscript.

Data sharing statement

Qualified researchers may request access to individual patient-level data through the clinical study data request platform (<http://www.clinicalstudydatarequest.com>). Further details on Roche's criteria for eligible studies are available here (<https://clinicalstudydatarequest.com/Study-Sponsors/Study-Sponsors-Roche.aspx>). For further details on Roche's Global Policy on the Sharing of Clinical Study Information and how to request access to related clinical study documents, see here (http://www.roche.com/research_and_development/who_we_are_how_we_work/clinical_trials/our_commitment_to_data_sharing.htm).

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Abbreviations: BFI, Brief Fatigue Inventory; CPI, checkpoint inhibitor; ES, effect size; HR, hazard ratio; (HR)QOL, (health-related) quality of life; i.v., intravenous/intravenously; ITT, intention-to-treat; LSM, least-squares mean; MDASI, MD Anderson Symptom Inventory; mRCC, metastatic RCC; PD-1, programmed cell death 1 protein; PD-L1, programmed death-

ligand 1; PFS, progression-free survival; PRO, patient-reported outcome; R-TKI, receptor tyrosine kinase inhibitor; VEGF, vascular endothelial growth factor.

Supporting Information

Additional Supporting Information may be found in the online version of this article:

Table S1. Patients' characteristics.

Table S2. Baseline PRO scores.